

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I G.HimaBindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Hima Bindu*

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Introduction

A smart retail inventory management system is developed using cloud computing to improve the management of products and sales in different retail stores. In this system, all branches of the retail chain are connected to a centralized cloud platform through the internet. Instead of maintaining separate records in every store, all inventory information is stored in one cloud database. This helps employees and managers access the latest information at any time. The system reduces manual work, improves accuracy, and helps the company provide better service to customers.

Client Devices and System Communication

The retail stores use different client devices such as Point of Sale (POS) systems, barcode scanners, billing machines, and manager dashboards. These devices are connected to the cloud server through the internet. Whenever a customer purchases a product, the cashier scans the barcode using the POS system. The POS immediately sends the product details to the cloud server. The server processes the request, updates the stock quantity, saves the sales record, and sends the updated information back to the billing system. This process takes only a few seconds and keeps inventory records accurate across all branches.

Role of Web APIs

The communication between client devices and cloud services is made possible through Web APIs. APIs act as a bridge between the retail application and the cloud server. Whenever the POS system requests product information or updates inventory, the API securely transfers the request to the cloud. The server processes the request and sends the response back to the client device. APIs also allow manager dashboards to display real-time sales reports, inventory status, and product availability. Because of APIs, different software applications can work together smoothly without any communication problems.

Cloud Storage and Data Management

Cloud storage is responsible for storing all important business information safely. It stores product details, supplier information, transaction records, invoices, customer purchase history, stock reports, and daily sales data. Since the data is stored in the cloud, employees from different branches can access the same updated information

whenever required. Cloud storage also provides automatic backup and data replication. Even if one server or local computer fails, another copy of the data is available, reducing the risk of data loss. As the business grows, additional storage space can be added easily without replacing existing hardware.

Platform Services and Business Analytics

The cloud platform provides different services that help improve business performance. One important service is analytics, where the system studies previous sales records and customer buying habits. Based on this information, the platform predicts which products may have higher demand during weekends, festivals, or seasonal sales. Managers can use these reports to order products before the stock becomes empty. The platform also generates reports showing sales performance, profit, product movement, and customer preferences. These reports help management make better business decisions and improve inventory planning.

Infrastructure, Scalability, and Performance

The inventory system runs on cloud infrastructure, which includes virtual machines, databases, networking services, and storage resources. Cloud infrastructure provides enough computing power to support many retail branches at the same time. During busy shopping seasons, customer traffic increases, and the cloud automatically allocates additional resources using auto-scaling. Load balancing distributes requests among different servers so that no single server becomes overloaded. This ensures that billing systems continue working smoothly even during peak business hours. Because cloud resources can be increased or decreased according to demand, the company only pays for the resources it actually uses.

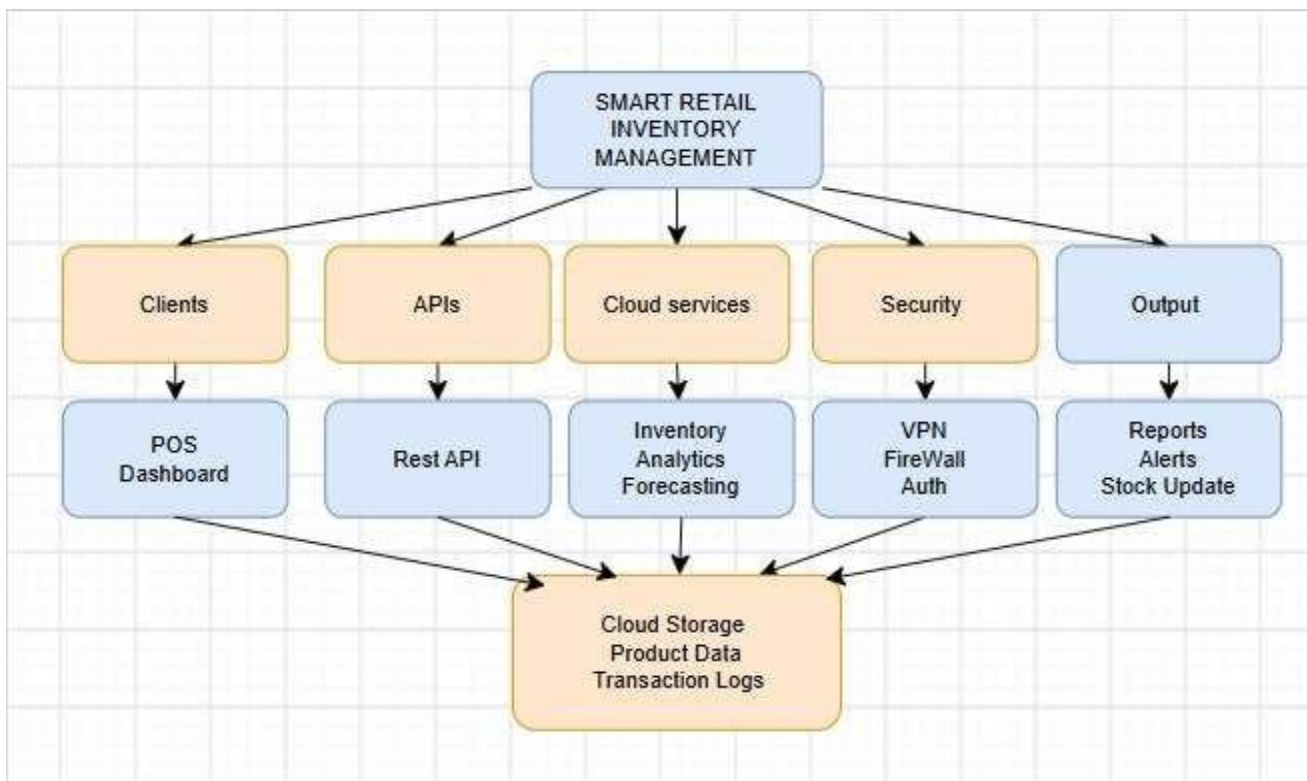
Security Features

Security is one of the most important parts of the retail inventory system because it stores valuable business and customer information. The company uses Virtual Private Networks (VPNs) to establish secure communication between retail stores and the cloud server. Firewalls protect the network by blocking unauthorized access and suspicious traffic. API gateways verify every request before allowing access to cloud services. User authentication ensures that employees log in using valid credentials before accessing the system. Role-Based Access Control (RBAC) gives different permissions to cashiers, managers, and administrators according to their responsibilities. Data encryption is also used during storage and transmission to protect sensitive information from cyber attacks.

System Workflow

The operation of the Smart Retail Inventory Management System begins when a customer purchases a product at the checkout counter. The POS system records the transaction and sends the purchase details to the cloud server through an API. The cloud application validates the request and updates the centralized inventory database by reducing the available stock quantity. Simultaneously, the transaction details are stored in cloud storage, and the analytics service updates sales reports and performance metrics. If the inventory level falls

below a predefined threshold, the system automatically generates alerts or purchase orders for suppliers. Managers can immediately view updated inventory levels and sales reports through cloud dashboards, ensuring real-time decision-making and efficient control.



2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Introduction

A cloud-based document collaboration system allows users to create, edit, store, and share documents through the internet. It is widely used in companies, educational institutions, and organizations where many people need to work on the same document. Since the application is hosted on the cloud, users can access their files using a web browser from a computer, laptop, tablet, or mobile phone. This makes collaboration easier because people can work from different locations without carrying physical files or installing special software.

Working of the Collaboration System

The system follows a client-server architecture. Users access the application through web browsers, which act as clients. Whenever a user opens or edits a document, the browser sends a request to the cloud server. The server processes the request, retrieves the latest version of the document, and displays it to the user. If changes are made, the updated information is immediately saved in the cloud. Since all users access the same cloud document, everyone sees the latest version without needing to exchange files through email or external storage devices.

Role of Web APIs and Real-Time Synchronization

Web APIs play an important role in connecting the client applications with the cloud services. Every action performed by a user, such as typing text, inserting images, or adding comments, is sent to the cloud through secure APIs. The cloud processes these changes and instantly updates the document for all users who are currently working on it. This real-time synchronization allows multiple users to edit the same document at the same time without creating different copies. APIs also support features such as document sharing, notifications, comments, and automatic saving, making collaboration smooth and efficient.

Cloud Storage and Version Control

Cloud storage is used to store all documents, presentations, spreadsheets, and other files safely. The documents are stored in distributed cloud servers, which means multiple copies of the same file are maintained in different locations. This prevents data loss even if one server fails. Another important feature is version control. Every time a user makes changes to a document, the system automatically saves a new version. If a mistake is made or important content is accidentally deleted, users can easily restore an earlier version. This feature improves reliability and protects important business documents.

Network Infrastructure and High Availability

The cloud platform uses a strong network infrastructure to provide fast and reliable access to users around the world. Load balancers distribute incoming requests among multiple servers so that no single server becomes overloaded. Content Delivery Networks (CDNs) help reduce network delay by delivering data from servers located closer to users. Auto-scaling automatically increases computing resources when many users access the system at the same time and reduces resources during low traffic. These technologies ensure that the collaboration platform remains available even during peak usage.

Security and Access Management

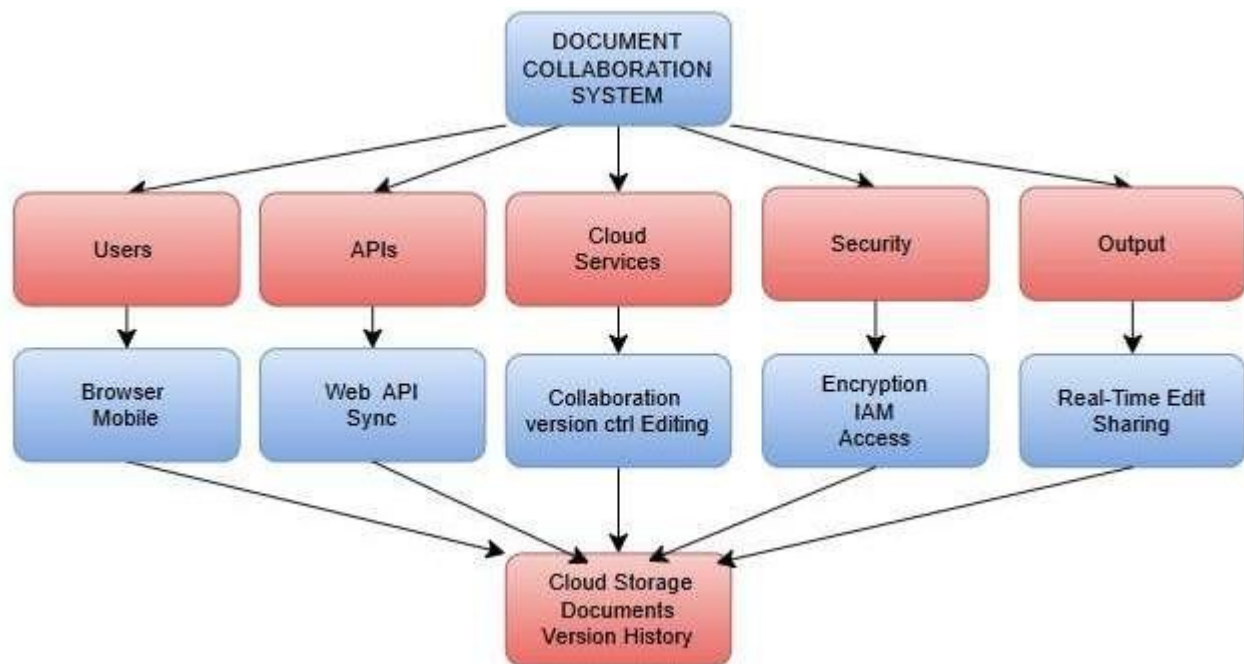
Security is very important because confidential company documents are stored in the cloud. The system uses HTTPS encryption to protect data while it is transferred between users and cloud servers. Identity management verifies the user's login credentials before granting access. Role-based access control allows document owners to decide whether other users can only view, comment, or edit a document. Some organizations also use multi-factor authentication, where users provide an additional verification code along with their password. These security measures protect sensitive information from unauthorized access and cyber attacks.

Benefits of the Cloud-Based Collaboration System

The cloud-based collaboration system provides many advantages to organizations. Employees can work together from different locations without facing communication problems. Documents are automatically saved, reducing the risk of losing important information. Real-time editing improves teamwork because multiple users can contribute simultaneously. Automatic backups and version control make document recovery easy if any errors occur. Since the system is cloud-based, organizations do not need to purchase expensive storage devices or maintain complex infrastructure. It also improves productivity by allowing users to access documents anytime and from any device with an internet connection.

System Workflow

The workflow begins when a user logs into the collaboration platform through a web browser. The user creates a new document or opens an existing one stored in the cloud. As the user edits the document, every change is immediately transmitted to cloud servers through Web APIs. The cloud application processes these updates, saves them in distributed cloud storage, records them in the version history, and synchronizes the latest changes with all other users currently editing the document. If another user joins the collaboration session, they instantly receive the latest version of the document. Throughout the process, encryption protects data transmission, while identity management and access permissions ensure that only authorized users can access or modify the document



3.Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Introduction

An online food ordering platform is a cloud-based application that allows customers to order food from different restaurants using a web browser or a mobile application. The entire system is hosted on the cloud, making it easy for customers, restaurants, and delivery partners to access the platform from anywhere. Customers can browse menus, place orders, make online payments, and track their deliveries in real time. Cloud computing helps the platform handle a large number of users while providing fast and reliable service.

Working of the System

The food ordering platform follows a client-server architecture. The clients are the web browsers and mobile applications used by customers. When a customer opens the application and selects food items, the request is sent to the cloud server. The server processes the request, checks the restaurant details, confirms the availability of the selected items, calculates the total bill, and sends the response back to the customer. After the order is confirmed, the restaurant receives the order details, and the delivery partner is notified to deliver the food. This entire process happens within a few seconds through the cloud platform.

Role of RESTful Web APIs

RESTful Web APIs play an important role in connecting the frontend applications with the backend cloud services. Every action performed by the customer, such as viewing the menu, searching for restaurants, placing an order, or making a payment, is carried out through APIs. The APIs transfer requests securely to the cloud server and return the required information quickly. They also help restaurants update menu items, prices, and order status in real time. Because of REST APIs, all parts of the system communicate smoothly and provide a better user experience.

Cloud Storage and Data Management

Cloud storage is used to store all important information related to the platform. It stores restaurant details, food menu images, customer profiles, delivery addresses, payment records, invoices, and order history. The cloud database automatically saves and backs up the information, reducing the possibility of data loss. Since

cloud storage is highly scalable, it can store a large amount of data even as the number of customers and restaurants increases. This allows the business to expand without worrying about storage limitations.

Cloud Infrastructure and Scalability

The platform uses cloud infrastructure consisting of virtual machines, databases, networking services, and managed Kubernetes clusters. Virtual machines host the application, while Kubernetes manages the containers that run different services such as order management, payment processing, and notifications. During busy hours like lunch, dinner, weekends, or festivals, the number of users increases rapidly. Auto-scaling automatically adds more computing resources to handle the increased traffic. Load balancing distributes customer requests among multiple servers, ensuring that the application remains fast and responsive without downtime.

Security Features

Security is essential because the platform handles customer information and online payments. Authentication tokens verify the identity of users before allowing access to their accounts. HTTPS encryption protects the communication between customers and the cloud server, ensuring that personal and payment information remains secure. Role-Based Access Control (RBAC) provides different levels of access to customers, restaurant owners, delivery partners, and administrators according to their responsibilities. Regular monitoring, secure payment gateways, and encrypted databases further improve the overall security of the platform.

Benefits of the Cloud-Based Food Ordering System

The cloud-based food ordering system offers many advantages to both customers and businesses. Customers can place orders from anywhere and receive instant confirmation and delivery updates. Restaurants can manage orders efficiently and update their menus without difficulty. Cloud computing provides high availability, automatic backup, and reliable performance even during peak hours. Auto-scaling and Kubernetes improve resource utilization and reduce operational costs. The platform also provides analytics that help restaurants understand customer preferences and improve their services.

System Workflow

The operation of the online food ordering platform begins when a customer accesses the application through a web browser or mobile app and logs into the system. The customer browses restaurant menus, selects food items, and places an order. The frontend application sends the order details to backend cloud services through RESTful APIs. The backend validates the authentication token, verifies menu

availability, calculates the total payment, and records the order in the cloud database. Menu images and invoices are stored securely in cloud storage, while confirmation notifications are sent to both the customer and the restaurant. Restaurant staff prepare the order, and delivery personnel receive delivery information through their mobile application. Throughout the process, Kubernetes manages application scalability, HTTPS secures communication, and RBAC ensures that each user accesses only authorized system functions.

